

FROM SENSOR TO CLOUD: A Plug-And-Play Approach Evolving

In a perfect world, connecting devices, machines and systems to the cloud should be as simple as connecting USB devices or PCIe expansion cards to a PC. Make the connection, open the cloud application, recognise devices and configure. You're now ready to make use of all your sensor data via a cloud connection, except that scenario is today nothing more than an application developer's dream.

In reality, because sensor data being gathered at the field level has any number of characteristics—such as size, structure, protocol, interface and transmission path—it has been virtually impossible to realise a direct cloud connection to all those various smart sensors with only a few clicks. The absence of local logic, required to process heterogeneous data into meaningful information, is a roadblock to making sense of incredibly diverse sensor data.

Yet the appeal of the cloud's promise to access Big Data and sophisticated analytics anytime, anywhere has application developers inspired to focus on local gateway technology to find a functional path toward the plug-and-play vision. Today, more powerful, evolved gateways, which can function either as dedicated devices or as a virtual part of a system, play a new role in receiving,

translating, processing and transmitting data as transparent information to the spectrum of cloud interfaces. This is enabled by the new cloud API for IoT gateways. It is essentially a middleware and glue logic solution to enable simple orchestration of wired and wireless sensor networks as well as embedded system configurations. The cloud API provides application-ready software modules that act as blueprints for original equipment manufacturers (OEMs) to develop their own applications, removing complexity and creating a smart path to connect all types of sensor networks to any cloud platform.

Amplifying the importance of gateways

Gateways are complex devices with excellent transcoding and decision-making capabilities. Using integrated logic, these collect, analyse and transcode sensor data and then determine whether it goes to the field, the cloud or perhaps another gateway. Their secure end-to-end encryption further allows them to structure and move data consistently, for example, enabling bidirectional communication with a specific cloud solution.

Enabled with the new cloud API, the IoT gateway communicates locally with intelligent sensors, now capable of processing and converting the received sensor data. Embedded driver modules (EDMs) interface with hardware and third-party expansion cards, providing the glue logic that translates received data into the semantics of the application-specific IoT gateway logic. This sensor engine, with EDM modules incorporated in its structure, is among the first software components to be standardised as a cloud function module. Its critical value is in moving data from local sensors to a generic middleware, independent of protocols.

Data is transmitted into the cloud via predefined interfaces and decision processes, at the same time normalising data measurements and checking that received data is meaningful. By using transparently defined modules and function blocks, identical logic interfaces can be used in each application.



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Standardised building blocks are becoming a new imperative in cloud system and application development

Fig. 1: The cloud API environment

